

Faizan Farhad

Aspiring AI/ML Engineer

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PROFILE

BCA student specializing in Large Language Models (LLMs) and Agentic AI workflows. Experienced in architecting automated data analysis pipelines, designing dynamic system prompts, and **orchestrating** multi-agent systems via LangChain. Adept at building scalable, AI-driven applications that bridge data validation with generative intelligence.

EDUCATION

BCA | CPJ College
2024 – 2028(Expected) | Narela, Delhi

SKILLS

- **Languages:** Python (Proficient), Java (Familiar)
- **Machine Learning:** Scikit-Learn, Predictive Modeling, Classification Modeling
- **Data Stack & Tools:** Pandas, NumPy, Git & GitHub, Data Visualization, APIs
- **Agentic AI & LLMs:** LangChain, LLM Workflows, Prompt Engineering
- **NLP:** Text Preprocessing, Feature Engineering, Tokenization, BPE
- **Deep Learning:** PyTorch, CNN, Neural Networks, Transformer Encoders, Attention Mechanisms

LANGUAGES

- Hindi
- English

COURSES

AI/ML | Daksh Gurukul (IIT Guwahati)

June 2025 – Oct 2026

- **Acquiring** deep theoretical foundations and practical applications in AI/ML principles, deep learning frameworks, and predictive modeling techniques to solve real-world problems.

PROJECTS

Local RAG Research Assistant

July 2026 – Aug 2026

- Engineered a fully local **Retrieval-Augmented Generation (RAG) chatbot** for research papers using **FAISS, Sentence Transformers, and Qwen 2.5 1.5B**, enabling document-grounded question answering without external API dependencies.
- Built an end-to-end document pipeline covering **PDF/Arxiv ingestion, text extraction, cleaning, token-based chunking, embedding generation, vector indexing, top-k retrieval, and LLM-based answer generation.**
- Implemented **chunk-level source citations** and a retrieval evaluation framework using **Recall@k, MRR, nDCG@k, Context Precision, Faithfulness, F1, and Exact Match (EM)** to assess retrieval and answer quality.
- Developed an interactive **Streamlit interface** with PDF uploads, Arxiv paper scraping, conversational querying, and expandable source references; optimized local inference using **model caching and 4-bit quantization.**

Tech Stack: Python, FAISS, Sentence Transformers, Qwen 2.5, Hugging Face, Streamlit, PyMuPDF, Selenium

Transformer Encoder From Scratch – Sentiment Analysis

Apr 2026 – May 2026

- Implemented a **Transformer Encoder-based sentiment classifier in PyTorch**, including token embeddings, positional encoding, multi-head self-attention, feed-forward networks, residual connections, and layer normalization.
- Built an end-to-end **encoder-only architecture** with configurable `d_model`, attention heads, encoder layers, and feed-forward dimensions, followed by mean pooling and a linear classification head.
- Developed and trained the model on the **sentiment dataset from kaggle**, implementing the training and evaluation pipeline with **AdamW, learning-rate scheduling, weight decay, and gradient clipping.**
- Analyzed training and validation performance to identify **overfitting**, with configurable dropout, model capacity, regularization, and optimization parameters for experimentation.

Tech Stack: Python, PyTorch, NumPy, Pandas, Hugging Face Datasets, Scikit-Learn